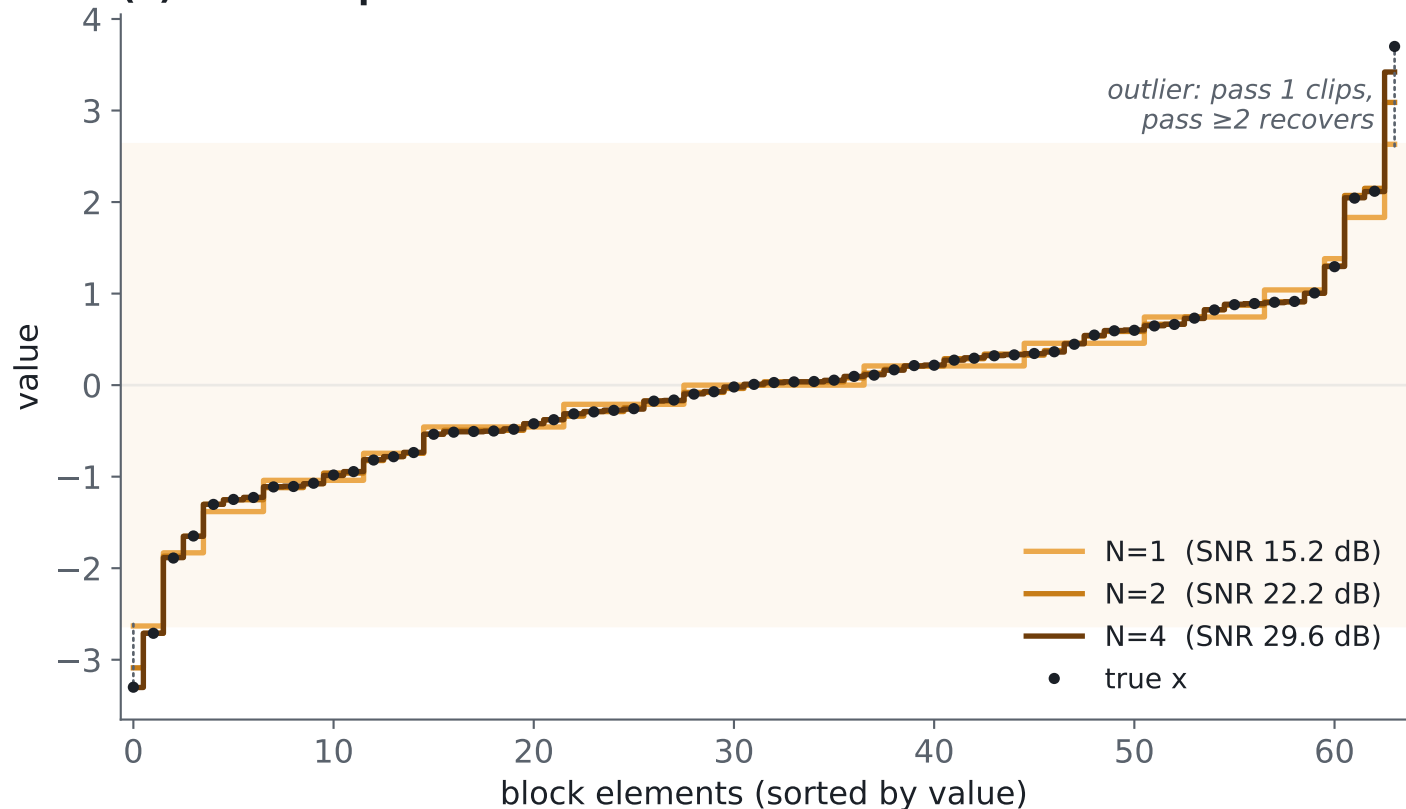


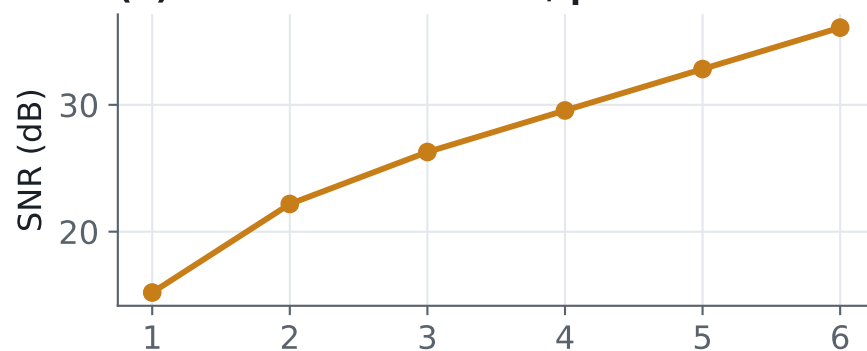
GF4 multi-pass residual quantization: $q \leftarrow q + \text{GF4}(x - q)$

the 4-bit GF4 grid is re-used every pass — no added precision, no second datapath

(a) Residual passes refine the reconstruction



(b) SNR climbs ~ 4.2 dB / pass



(c) Leftover error decays geometrically

